

Module 01: Systems — What Are Living Systems?

Course: Active Inference 101 | **Unit:** Cognitive Science | **Audience:** First-semester undergraduates

Learning Objectives

1. Define a **system** and distinguish open systems from closed systems.
2. Explain what a **Markov Blanket** is and why it matters for understanding organisms.
3. Describe how living systems maintain themselves far from equilibrium through homeostasis and allostasis.
4. Recognize systems-level thinking as a foundation for understanding minds.

Introduction

Before we can understand how minds work, we need to understand what kind of *thing* a mind is. Active Inference begins with a radical claim: **the same principles that keep a cell alive also explain how brains think**. But to see this, we first need to understand systems — organized wholes that maintain themselves by exchanging matter and energy with their environment.

Key Insight: Active Inference doesn't start with brains, neurons, or consciousness. It starts with the question: *What does any living thing need to do to keep existing?* The answer — maintain boundaries, predict threats, and act to survive — turns out to scale all the way from bacteria to human cognition.

Key Concepts

1. What Is a System?

A **system** is a set of interacting parts that together produce behavior that none of the parts produce alone. Examples:

- A **cell**: Organelles interact to metabolize, divide, and respond to signals

- A **university**: Students, professors, buildings interact to produce education
- An **ecosystem**: Species interact through food webs, competition, and symbiosis
- A **smartphone**: Hardware and software interact to run applications

Notice: the system is not just the sum of its parts. Cut a frog in half and you don't get two half-frogs — you lose the system. This is called **emergence** — the whole has properties that the parts lack.

2. Open vs. Closed Systems

- **Closed system**: Exchanges no matter or energy with the environment (very rare — almost nothing is truly closed). A sealed thermos is approximately closed.
- **Open system**: Exchanges both matter and energy with the environment. All living things are open systems — they need food, oxygen, and waste removal to survive.

The critical consequence: living systems must constantly *work* to maintain their organization against the constant tendency of the environment to dissolve them. This work is what Active Inference models.

3. Markov Blankets — The Boundary of Self

Every living system has a **boundary** that separates inside from outside. In Active Inference, this boundary is formalized as a **Markov Blanket** — a statistical concept that divides the world into:

- **Internal states**: What's inside the system (e.g., your body's internal organs, a cell's cytoplasm)
- **External states**: What's outside the system (e.g., the weather, predators, food sources)
- **Sensory states**: How the outside affects the boundary (e.g., your skin sensing temperature, a cell's receptors detecting glucose)
- **Active states**: How the inside affects the boundary (e.g., your muscles moving your body, a cell's flagella propelling it)

Why "blanket"? Just as a blanket wraps around you to separate you from the cold, the Markov Blanket wraps around the internal states, separating them from the external world. The crucial point: **internal states and external states are conditionally independent given the blanket**

states — meaning the inside can only "know about" the outside through its sensory and active states.

Real-World Example — Your Skin: Your skin is part of your Markov Blanket. It senses the external world (touch, temperature, pain) and acts on it (sweating, flushing, hair bristling). Your internal organs never directly contact the external world — everything is mediated by the blanket. This is not just a metaphor; it's the mathematical foundation of Active Inference.

4. Homeostasis and Allostasis

Living systems maintain themselves through two complementary strategies:

Homeostasis (reactive regulation): Detecting a deviation from a setpoint and correcting it.

- You overheat → you sweat (reactive response to current state)
- Your blood sugar drops → you feel hungry (reactive response)
- A thermostat detects temperature change → turns on heating/cooling

Allostasis (predictive regulation): Anticipating future needs and preparing in advance.

- You eat breakfast before your blood sugar drops (anticipatory)
- You put on a coat before going outside (anticipatory)
- Bears fatten up before winter (anticipatory)

The critical difference: homeostasis reacts to *current* prediction errors; allostasis acts on *predicted future* prediction errors.

Key Insight — From Thermostats to Thinkers: Homeostasis is what a thermostat does. Allostasis is what a brain does. The leap from homeostasis to allostasis — from reacting to predicting — is the leap that makes cognition possible. Active Inference is the formal framework that captures this leap.

5. Far from Equilibrium

Living systems are **far-from-equilibrium** systems — they maintain a specific, highly organized state that would rapidly decay without constant energy input. A dead body is at

equilibrium with its environment; a living body actively resists equilibrium (dissolution, decay).

This constant resistance to dissolution is what Active Inference formalizes as **free energy minimization** — the system acts to keep itself in its characteristic states, resisting the second law of thermodynamics' drive toward disorder.

6. Connecting to Active Inference

This module's concepts are the foundation of everything that follows:

Systems Concept	Active Inference Concept	Module Connection
Open system	Agent exchanging with environment	Module 02: Agents
Markov Blanket	Statistical boundary of any self	Module 02, 03
Homeostasis	Reactive belief updating	Module 03: Perception
Allostasis	Predictive action and planning	Module 08: Planning
Far from equilibrium	Free energy minimization	Module 04: Cognition
Emergence	Complex behavior from simple rules	Module 07: Communication

Summary

A system is a set of interacting parts whose collective behavior is more than the sum of its parts. Living systems are open, far-from-equilibrium systems that maintain themselves through homeostasis (reactive) and allostasis (predictive). The Markov Blanket formalizes the boundary between self and world — everything the system knows about the outside world comes through its sensory and active states. These concepts form the foundation for understanding agents, perception, and cognition in Active Inference.

Further Reading

- Friston, K. (2013). Life as we know it. *Journal of the Royal Society Interface*, 10(86), 20130475.
- Clark, A. (2016). *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*. Oxford University Press. (Chapter 1)
- Kirchhoff, M., Parr, T., Palacios, E., Friston, K., & Kiverstein, J. (2018). The Markov blankets of life: autonomy, active inference and the free energy principle. *Journal of the Royal Society Interface*, 15(138), 20170792.
- Nicolis, G. & Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems*. Wiley.